Guide Dogs

Indoor navigation for blind and low-vision impaired

SE 4351 Requirements Engineering • Fall 2026
Project Phase I • Interim Presentation
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Team 5

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Problem

Current visual impairment assisting software applications are limited in the accessibility, and comfort of the user. Some solutions force users to purchase third-party Hardware to pair with the application, adding more objects to carry around. This makes the visually impaired have to add yet another device when trying to improve their safety, especially when in close spaces.

As-Is & To-Be Scenarios (1 of 3): Wrong Room

AS-IS



1. Heads to ECSS 2.410, class in 3 minutes



2. Reads a sign that belongs to the next door

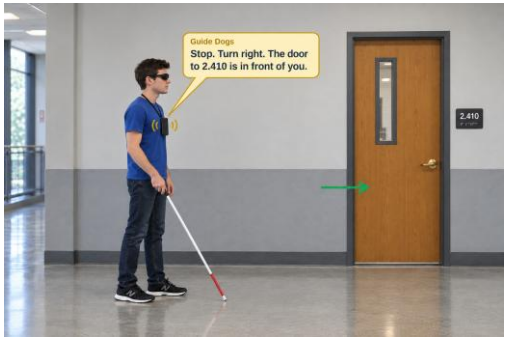


3. Walks into 2.412 in the middle of a lecture

TO-BE



1. Says the room; the app plans the route



2. Told to stop and turn at the right door



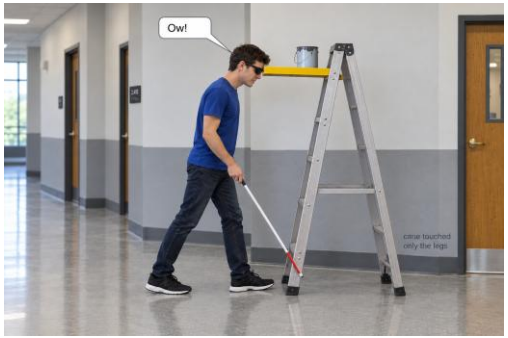
3. Right room, on time

As-Is & To-Be Scenarios (2 of 3): Obstacle Collision

AS-IS



1. Walks toward a ladder left in the hallway



2. The cane finds the legs, not the shelf



3. Hurt, and unsure the path ahead is clear

TO-BE



1. Chest camera spots the ladder 4 steps out



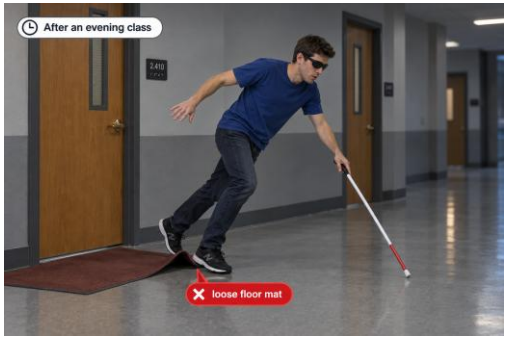
2. Steps left, around the ladder



3. Reaches class safely

As-Is & To-Be Scenarios (3 of 3): Fall With No One Around

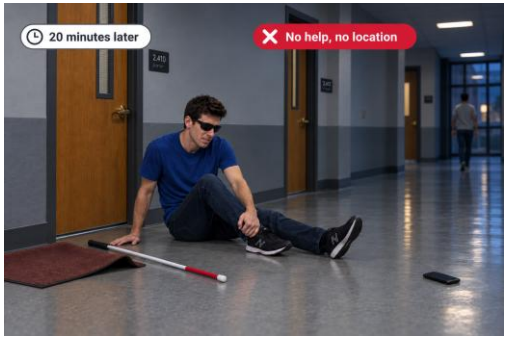
AS-IS



1. Trips on a loose mat after an evening class



2. Phone out of reach, nobody nearby



3. Waits 20 minutes on the floor

TO-BE



1. The app detects the fall and checks in



2. Alex picks who to call; location is sent



3. Help arrives within minutes

Stakeholders

OF
Part of the system

Caretaker
Gets emergency alerts

UTD Police
Answers emergency calls on campus

911 Dispatch
Backup if there is no answer

BY
Who builds or operates

Group 5
Designs and builds the app

FOR
Who it serves or affects

Blind and Low-Vision Users
Main users who need to get around ECSS safely

UTD Student AccessAbility
Wants Blind & Low-vision students to move around on their own

Bystanders
May be seen by the camera in hallways

Tools, Standards, & Templates

Figma

Google Docs

Google Slides

Gemini



Presentation Template



pg-WRS Template

Issues with the World/Domain (II.1)

ID	Issue in the preliminary definition	Type	Decision
DI1	"Indoors" is undefined: which buildings and floors, and what counts as "connected"?	*Ambiguity, incompleteness*	Phase 1 covers the ECSS 1st floor. Floors are stored as map data, so more can be added without new features.
DI2	No map or positioning method, and GPS does not work reliably indoors.	*Incompleteness (missing domain assumption)*	The camera reads room-number signs against a preloaded floor plan; step counting tracks the user between signs.
DI3	The caretaker must "come to aid," but is location shared all the time?	*Ambiguity*	Location is shared only during an emergency or when the user asks for help.
DI4	"Blind" is not scoped: total blindness, low vision, blindness with hearing loss.	*Ambiguity*	Add low-vision support and a vibration-only mode with four distinct cues.

Each issue lists its options, decision and rationale in the pg-WRS, section 2.1.

Issues with Functional Requirements (II.2)

ID	Issue in the preliminary definition	Type	Decision
FRI1	Emergency calls: nobody is named to call, and losing the location is treated as an emergency.	*Incompleteness, unsoundness*	The user picks caretaker, UTD Police or 911; no answer in 30 s calls 911 and notifies the caretaker. Being lost only offers the caretaker.
FRI2	Obstacle detection: which obstacles, how far ahead, and how can a pocketed phone see?	*Incompleteness*	Floor to head height, including drop-offs, at least 4 steps ahead, via a chest-lanyard camera. Moving people wait for Phase 2.
FRI3	Distance mixes minutes and steps, and step length differs per person.	*Ambiguity*	The user picks steps, feet or seconds (default steps); stride is calibrated at setup.
FRI4	No FR finds the starting location or reroutes, and the list ends in "……."	*Incompleteness*	Detect the start from the nearest sign and confirm it; reroute when off course; every feature gets a numbered FR.

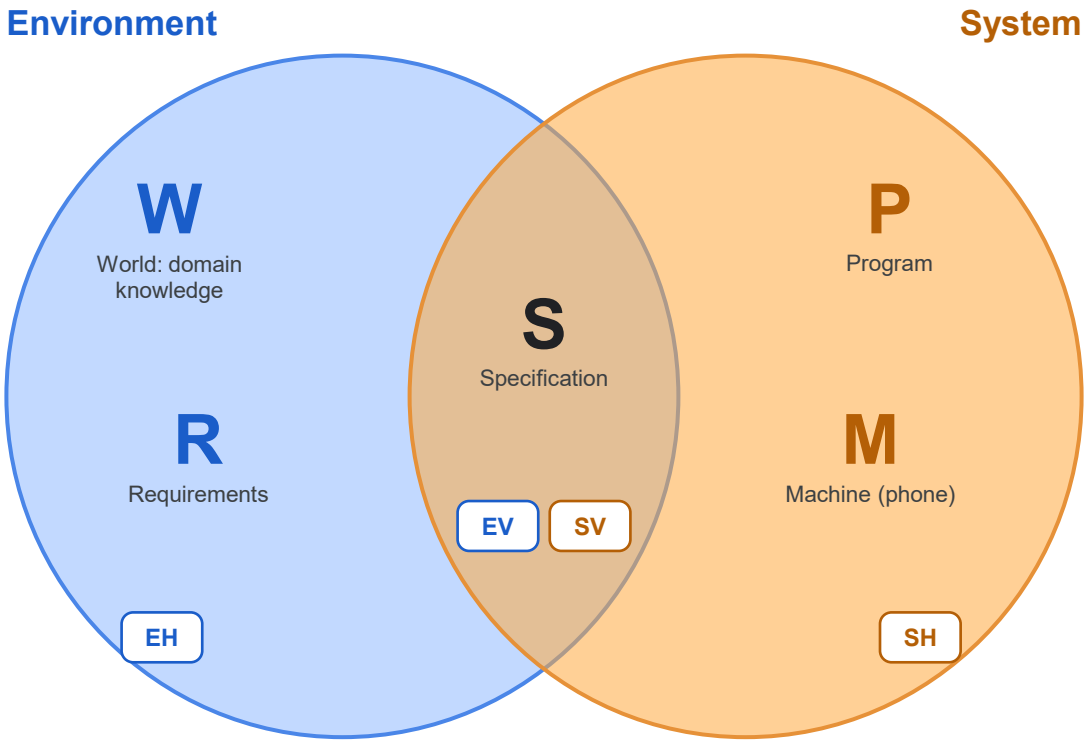
Each issue lists its options, decision and rationale in the pg-WRS, section 2.2.

Issues with Non-Functional Requirements (II.3)

ID	Issue in the preliminary definition	Type	Decision
NFRI1	Safe, fastest and most comfortable can each pick a different route.	*Conflict*	Safest by default; saying "I'm running late" switches to the fastest route that still avoids known hazards.
NFRI2	Customizable and extensible say "should"; every other NFR says "shall."	*Inconsistency*	All NFRs use "shall." Our design depends on customization, so it cannot be optional.
NFRI3	"Usable" and "ubiquitous" cannot be tested as written.	*Ambiguity*	Usable: every function by voice, works with VoiceOver and TalkBack, learnable in 10 minutes. Ubiquitous: iOS and Android, offline on mapped floors.
NFRI4	No privacy or battery NFR, and two NFRs end in "etc."	*Incompleteness*	Add privacy (NFR8) and battery at most 15% per 30 minutes (NFR9); replace each "etc." with a complete list.

Each issue lists its options, decision and rationale in the pg-WRS, section 2.3.

pg-WRS Reference Model: Environment and System



Shared phenomena (EV, SV) live in the overlap.

EH Environment, hidden from the system

Alex's destination before it is spoken; an obstacle around a corner; whether the caretaker is free

EV Environment, visible to the system

Alex's voice commands and replies; Alex's steps; obstacles in camera view; room-number signs; a fall

SV System, visible to the environment

Spoken directions; vibration cues; emergency calls; location messages to the caretaker

SH System, hidden from the environment

Route calculation; the stored floor plan; stride and position estimates

Reference Model Example: Obstacle Warning

S

Specification

When the camera detects an object in the walking path within 4 steps, the system shall vibrate and say which way to move.

+

D

Domain assumptions (W)

The phone is worn facing forward on the chest lanyard. The hallway is lit well enough for the camera. Alex perceives the warning and follows it.

⊨

R

Requirement

Alex does not collide with obstacles in the hallway.

S, D ⊨ R The specification and the domain assumptions together satisfy the requirement. If an assumption fails (a dark hallway, the phone in a pocket), R can fail even though S holds.

Scenario 2 → **FRI2** → **FR7, FR11** → **NFR1 safety**

Traceability for this example

Goal

The goal for Guide Dog is to provide a **seamless** integration with current assistive devices to further provide **safety** for the visually impaired when indoors.

- Provide Safety when navigating indoors
- Fast response time for Emergencies
- Give accurate routes
- Integrates with other assistive devices seamlessly
- On-Demand assistance

Improved Understanding of the World/Domain & Stakeholders

ID	Improved Domain Description
ID1	For this smartphone app, the domain will be the ECSS first floor, including classrooms, offices, washrooms, lounges, elevators, stairs, and laboratories.
ID2	The app will use the existing floor plan of the building and room signs to determine the user's current position within the building.
ID3	The application will share the user's location with their caretaker only during an emergency, which occurs when the user has fallen or asks for help.
ID4	The primary stakeholders are blind and low-vision users, including users who are also deaf, who need to navigate indoors.

Improved Understanding of Functional Requirements

ID	Improved Functional Requirement Description
FR2	The system shall instruct the user to walk a specified distance, defined in time, or steps (defined by user in calibration).
FR4	The system shall automatically prompt the user to choose which emergency contact to notify (defaulting to emergency services), then places calls to emergency services and send text messages to user-provided emergency contacts upon detecting a fall or when the system fails to determine the current location.
FR8	The system shall notify the user of their devices remaining battery percentage for every 10% charge used, with additional notifications when there is critical charge at 20% and 5% remaining.
FR14	The system shall assume its current location based on the context of its previous route and current locational data.

Improved Understanding of Non-functional Requirements

ID	Improved Non-Functional Requirement Description
NFR4	The system shall be fully usable without sight, with every function operable with voice commands and signaled with audio or haptic feedback.
NFR5	The system shall function reliably and consistency between across all supported indoor spaces without unexpected failures.
NFR8	The system shall temporarily and securely keep information collected through encryption, disposing of temporary information within 30 days.
NFR9	The system shall use an average of 5% battery life per hour of use.

Preliminary Prototype



Home
Voice-first: the whole screen is one button



Guidance
Step countdown in the user's chosen unit



Obstacle
Full-screen warning plus rapid vibration



Fall check
Choose who to call; 911 after 30 s

Preliminary User Manual

Quick start

1. Open Guide Dogs. Every screen is spoken, and VoiceOver and TalkBack work.
2. Choose a distance unit: steps, feet or seconds.
3. Calibrate your stride by walking 10 normal steps.
4. Practice the four vibration cues.
5. Add your caretaker as an emergency contact (a sighted helper can do this).
6. Clip the phone to the chest lanyard, camera facing forward.
7. Tap anywhere and say where to go: "Take me to ECSS 2.410."

Your location is shared only during an emergency or when you ask for help.

Vibration	Meaning
One long buzz	Stop
Two short	Turn left
Three short	Turn right
Rapid pulses until clear	Obstacle ahead

Command card

Say	What happens
"Take me to [room]"	Starts a route
"Where am I?"	Tells you the nearest room
"What are my options?"	Lists the available routes
"Repeat"	Repeats the last instruction
"I'm running late"	Fastest hazard-free route
"Pause" / "Stop navigation"	Pauses or ends the route
"Help"	Offers to contact your caretaker
"I'm fine"	Cancels a fall alert
"Call caretaker" / "Call campus police" / "Call 911"	Places that call and sends your location

Phase 1 draft; caretaker setup screens come in Phase 2.

Questionnaire

1. **How would you like to receive real-time obstacle alerts?**
    a. Sound
    b. Vibration
    c. Both
2. **If you fall, who would you like to be notified first?**
    a. Caretaker (selected from your contacts)
    b. UTD Police
    c. 911
3. **How do you want route distance to be measured?**
    a. Number of steps remaining
    b. Feet remaining
    c. Seconds until reaching destination (calculated from your stride)
4. **When using the application, would you rather hold your phone or wear it on a lanyard?**
    a. Hold phone
    b. Wear on lanyard
    c. Other: _____

Traceability Matrix

Trace ID	Need / Finding	Domain Assumption	Functional Req (FR)	Non-Functional (NFR)	System Feature
TR-01	Hands-free way to specify destinations without touch screens	User speaks clearly; mic is unobstructed	FR1 / FR3: Voice destination input and stride-calibrated guidance	NFR3 / NFR4: Voice-first usability, high contrast, ≤10 min learning curve	Home & Guidance: Single full-screen tap-to-talk
TR-02	Cane misses elevated/hanging obstacles (head height)	Forward-facing chest lanyard camera, lit hallways	FR7 & FR11: Object detection within 4 steps + haptic alerts	NFR1: Safe navigation, hazard-free routing	Obstacle Screen: Yellow warning + rapid pulses
TR-03	User falls in empty hallway and phone flies out of reach	Accelerometer detects impact & lack of movement	FR6/8/12: Fall check-in, 30s countdown, auto-dial emergency	NFR8 / NFR1: Safety & privacy (emergency location only)	Fall Alert Screen: Red 30s auto-escalation screen
TR-04	Battery drainage leaves blind users stranded	Sensors optimize duty cycles during steady paths	FR8: Proactive voice alerts at low battery (20%, 5%)	NFR9: Power consumption ≤ 5% per hour	System Service: Background battery guard daemon

Requirements Creeping Rate

Requirements Creeping Rate (RCR) Estimate
Estimated Creeping Rate: 15% – 20% between Phase 1 and Phase 2.

Primary Drivers of Scope Creep:

- **Dynamic Obstacle Tracking (Phase 2):**
  Expanding vision algorithms from static obstacles (ladders, carts) to dynamic obstacles (walking pedestrians, opening doors).
- **Multi-Floor / Multi-Building Expansion:**
  Moving beyond the ECSS 2nd floor testbed to elevators, stairwells, and skybridges across adjacent buildings.
- **Caretaker Configuration UI:**
  Moving from phone voice commands to an external companion portal for family members/caretakers.

Creep Management & Control Strategy:

- **Baseline Scope Freeze:**
  Lock the core indoor navigation and fall-detection specification ($S, D \vdash R$) for the ECSS 2nd floor before adding new sensors.
- **Extensibility Architecture (NFR-Extensible):**
  Modular map and sensor interfaces allow floor updates without rewriting navigation logic.
- **Change Control Filter:**
  New feature requests are deferred to Phase 2 unless they directly resolve a life-safety hazard (NFR1).

Why our Team's Product is believed to be the Best

Full voice control ⇒ Hands-Free Navigation!

Accessible by Design ⇒ Built for Independence!

Safety features ⇒ Help When It Matters!

Real-time Hazard detection ⇒ More than Directions!

Revision History

Date	Author(s)	Change
09/15/26	Kyle	Issues w/ world, domain, functional and nonfunctional reqs
09/16/26	Kyle	Prototype, As-Is, To-Be
09/17/26	Luis	Problem and Goal
09/17/26	Quincy	Tools, standards, and templates
09/17/26	Dylan	Improved understanding of functional and non-functional requirements
09/18/26	Cynthia	Questionnaire
09/18/26	Adam	Requirements creeping rate, traceability matrix
09/20/26	Grace	Improved understanding of world and domain
09/27/26	Aaron	Stakeholders
09/29/26	Julia	Why our product is believed to be the best